A strict upper bound for three-point weighted selection in planar vector-valued regression

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Abstract

We prove $F_{\text{weighted}}(2,2,3) < 11/6$ for minimum-Frobenius-norm least-squares regression with two input and two output coordinates, improving the upper bound $15/8$. More precisely, a universal $\varepsilon_0 > 0$ satisfies $\mathcal{T}_3 \le (5/6 - \varepsilon_0)E$ for every finite residual frame. The proof combines an exact scalar certificate, rigidity of equality in planar selection, and a compactness argument that controls disappearing weights. Together with the known lower bound, this gives $13/8 \le F_{\text{weighted}}(2,2,3) < 11/6$. No numerical value of ε_0 is claimed.

1 Problem and result

Weighted data selection for regression was proposed by Hanneke, Moran, Shlimovich and Yehudayoff [1, 2]. Zhang [3], Theorem 6.4, established $13/8 \le F_{\text{weighted}}(2,2,3) \le 15/8$ and reduced the conjectured value $13/8$ to a finite planar moment problem. We use Zhang's residual-frame formulation, interpolation hulls, star localization and seven-atom reduction. An exact scalar certificate, rigidity of equality and compactness yield a strict uniform improvement over all finite datasets. The conjectured value $13/8$ remains open.

For a finite sequence $D = ((x_i, y_i))_{i=1}^N$ in $\mathbb{R}^2 \times \mathbb{R}^2$, $N \ge 1$, let $L_D(W) = N^{-1} \sum_i \|Wx_i - y_i\|^2$ and $L_D^* = \min_W L_D(W)$. For nonnegative weights c of sum one, let $\widehat{W}_c$ be the minimum-Frobenius-norm minimizer of $\sum_i c_i \|Wx_i - y_i\|^2$. Then $F_{\text{weighted}}(2,2,3)$ is the supremum over datasets of

$$\frac{\inf_{c_i \ge 0,\, \sum_i c_i = 1,\, |\operatorname{supp} c| \le 3} L_D(\widehat{W}_c)}{L_D^*},$$

with $0/0 = 1$ and positive-over-zero equal to $+\infty$. The infimum, including limits of full-rank selections, is essential.

A residual frame is a finite collection $p_i > 0$, $u_i, w_i \in \mathbb{R}^2$ with

$$\sum_i p_i = 1, \quad \|u_i\|^2 = 2, \quad \sum_i p_i u_i u_i^\top = I_2, \quad \sum_i p_i w_i u_i^\top = 0. \tag{1}$$

Put $E = \sum_i p_i \|w_i\|^2$ and $R = \sum_i p_i w_i w_i^\top$. For independent u_i, u_j, let $G_{ij} u_i = w_i$, $G_{ij} u_j = w_j$. Using Frobenius distance, define

$$\mathcal{T}_3 = \min_{\substack{2 \le |S| \le 3 \\ S \text{ contains an independent pair}}} \operatorname{dist}\left(0, \operatorname{conv}\{G_{ij} : i, j \in S,\ \det(u_i, u_j) \ne 0\}\right)^2.$$

Tightness guarantees an independent pair.

Theorem 1. *There exists a universal $\varepsilon_0 > 0$ such that every residual frame satisfies $\mathcal{T}_3 \le (5/6 - \varepsilon_0)E$. Consequently $13/8 \le F_{\text{weighted}}(2,2,3) \le 11/6 - \varepsilon_0 < 11/6$.*

2 Planar selection and the scalar certificate

Lemma 2. *Let $v_j \in \mathbb{R}^2$ have positive probabilities p_j and mean zero. For any real 2×2 matrix A with $\operatorname{tr} A = 1$, some convex combination v of at most two of these points satisfies*

$$\|v\|^2 \le \sum_j p_j \|Av_j\|^2.$$

Proof. Minimize the right-hand linear objective over centered probability vectors on the fixed points, and choose a minimizer q of least support. If the columns $(1, v_j)$ on its positive support are dependent, a supported kernel vector has both signs. Small perturbations in either direction remain feasible, so optimality forces zero objective slope. Continuing until one weight vanishes contradicts least support. Hence at most three affinely independent points remain and their objective is no larger than the original. If at most two remain, their centered mean itself is zero.

Otherwise let $S_q = \sum_j q_j v_j v_j^\top$. It is positive definite: a null direction would place all three points on a line. Put $n_j = S_q^{-1} v_j$, choose i maximizing $r = \|n_i\|^2 > 0$, and set

$$\lambda_j = q_j \left(1 - \frac{n_j^\top n_i}{r}\right).$$

Cauchy–Schwarz and maximality give $\lambda_j \geq 0$. Centering gives $\sum_j \lambda_j = 1$, and $\lambda_i = 0$. Since $\sum_j q_j v_j n_j^\top = I_2$,

$$v = \sum_j \lambda_j v_j = -\frac{n_i}{r}, \qquad \|v\|^2 = \frac{1}{r}.$$

Finally, writing $C_q = \sum_j q_j \|A v_j\|^2$,

$$1 = \sum_j q_j n_j^\top A v_j, \qquad 1 \leq \left(\sum_j q_j \|n_j\|^2\right) C_q \leq r C_q.$$

Thus $\|v\|^2 \leq C_q$, proving the assertion. □

For positive semidefinite S with $\operatorname{tr} S > 0$, the matrix $A = \operatorname{adj} S / \operatorname{tr} S$ has trace one, and in dimension two

$$\operatorname{tr}(ASA^\top) = \frac{\det S}{\operatorname{tr} S}. \tag{2}$$

This identity holds also when S is singular. If a covariance has trace zero, all its positive-weight points vanish.

Fix an anchor i and a unit vector $n_i \perp u_i$. Put

$$M_i = \tfrac{1}{2} w_i u_i^\top, \quad b_{ij} = u_j^\top n_i, \quad c_{ij} = \tfrac{1}{2} u_i^\top u_j, \quad \xi_{ij} = \frac{w_j - c_{ij} w_i}{b_{ij}} \quad (b_{ij} \neq 0).$$

Because $u_j = c_{ij} u_i + b_{ij} n_i$,

$$G_{ij} = M_i + \xi_{ij} n_i^\top, \qquad \|M_i + \xi n_i^\top\|_F^2 = \tfrac{1}{2}\|w_i\|^2 + \|\xi\|^2.$$

The cross term vanishes since $u_i^\top n_i = 0$. The positive star weights $P_{ij} = p_j b_{ij}^2$ satisfy

$$\sum_{b_{ij} \neq 0} P_{ij} = 1, \qquad \sum_{b_{ij} \neq 0} P_{ij} \xi_{ij} = 0.$$

Indeed these are respectively $n_i^\top I_2 n_i = 1$ and the difference between the cross-moment $\sum_j p_j w_j u_j^\top n_i = 0$ and $w_i u_i^\top I_2 n_i / 2 = 0$.

Define

$$L_i = \sum_{b_{ij} = 0} p_j (w_j - c_{ij} w_i)(w_j - c_{ij} w_i)^\top \succeq 0, \qquad H_i = R + \tfrac{1}{2} w_i w_i^\top.$$

The star covariance is

$$S_i = \sum_{b_{ij} \neq 0} P_{ij} \xi_{ij} \xi_{ij}^\top = H_i - L_i. \tag{3}$$

To check it, expand the same sum over all j. The two cross terms vanish because $\sum_j p_j c_{ij} w_j = 0$, and $\sum_j p_j c_{ij}^2 = 1/2$. Subtracting the parallel terms yields (3).

Apply Lemma 2 with $A_i = \operatorname{adj} H_i / \operatorname{tr} H_i$; its denominator $E + \|w_i\|^2/2$ is positive for $E > 0$. PSD monotonicity and (2) give a two-neighbor convex mean with

$$\|\xi\|^2 \leq \operatorname{tr}(A_i S_i A_i^\top) = \frac{\det H_i}{\operatorname{tr} H_i} - \operatorname{tr}(A_i L_i A_i^\top).$$

Consequently, for every anchor,

$$\mathcal{T}_3 \le \frac{\|w_i\|^2}{2} + \frac{\det(R + w_i w_i^\top/2)}{E + \|w_i\|^2/2} - \operatorname{tr}(A_i L_i A_i^\top). \tag{4}$$

This involves at most three atoms. Dropping the last, nonnegative term is valid but potentially loses information.

Normalize $E = 1$ and rotate output coordinates so $R = \operatorname{diag}(r, 1-r)$. Set $a_i = w_{i1}^2$, $b_i = w_{i2}^2$, and $s = a + b$. The first two terms of (4) are

$$g_r(a,b) = \frac{s}{2} + \frac{r(1-r) + ((1-r)a + rb)/2}{1 + s/2}, \qquad \sum_i p_i a_i = r, \quad \sum_i p_i b_i = 1 - r.$$

Put $\delta = 2r - 1$, $x = 1 + 3\delta/10$, $y = 1 - 3\delta/10$, $P = xy$, $Q = 7/3 - 21\delta^2/100$, and $h = P - ay - bx$. Since

$$\sum_i p_i h(a_i, b_i) = \frac{21}{100}\delta^2 \ge 0,$$

some positive-weight atom has $h \ge 0$. Define

$$F(a,b) = s^2 + \frac{4}{3}s - \delta(a-b) - \frac{7}{3} - \delta^2.$$

Exact expansion gives

$$PF = -\frac{121}{100}\delta^2 P - h(Q + ax + by) - ab(x-y)^2, \tag{5}$$

$$4(1 + s/2)(g_r(a,b) - 5/6) = F(a,b). \tag{6}$$

Here $x, y, P, Q > 0$ for $|\delta| \le 1$. At the selected atom all terms on the right of (5) are nonpositive. This proves $\mathcal{T}_3 \le 5/6$. Homogeneity gives $\mathcal{T}_3 \le 5E/6$ for general energy.

We also need its quantitative anisotropy consequence. At $h \ge 0$, $s \le P/\min(x,y) = \max(x,y) \le 13/10$. Thus (5)–(6) give

$$\mathcal{T}_3 \le \frac{5}{6} - \frac{11}{60}\delta^2. \tag{7}$$

The coefficient is $(121/100)/(4(1 + 13/20)) = 11/60$.

The scalar relaxation alone is sharp: the one-atom scalar distribution $r = a = b = 1/2$ has $g = 5/6$. It is not a valid tight frame. The strict improvement must therefore use geometry.

3 Equality cannot occur

Lemma 3 (Equality rigidity)**.** *Suppose the centered positive distribution has positive definite covariance S, put $C = \det S/\operatorname{tr} S > 0$ and $A = \operatorname{adj} S/\operatorname{tr} S = CS^{-1}$, and let d^2 be the minimum squared norm of a convex combination of at most two original points. If $d^2 = C$, then every original point satisfies*

$$\|Av_j\|^2 = C, \qquad \|S^{-1}v_j\|^2 = \operatorname{tr}(S^{-1}) = 1/C.$$

Proof. The original A-objective is C by (2). Minimize it as in Lemma 2. Its minimum C_q obeys $d^2 \le C_q \le C$, so all equal C. A minimizing least-support distribution must be a centered positive triangle, since otherwise $d = 0$.

In its planar proof all inequalities are equalities:

$$C = d^2 \le \frac{1}{r} \le \frac{1}{\sum_j q_j \|S_q^{-1} v_j\|^2} \le C_q = C.$$

Equality in the maximum step forces all three dual norms to equal $r = 1/C$. Equality in weighted Cauchy–Schwarz forces $Av_j = CS_q^{-1}v_j$ on the triangle. Its vertices span $\mathbb{R}^2$, hence $A = CS_q^{-1}$, and comparison with $A = CS^{-1}$ gives $S_q = S$. Each of its three costs $\|Av_j\|^2$ therefore equals C.

For an original point v_k not in the selected support, write its affine coordinates in the triangle (which need not be nonnegative): $v_k = \sum_{j=1}^3 \beta_j v_j$, $\sum_j \beta_j = 1$. For sufficiently small $t > 0$, put weight t on v_k and change triangle weights to $q_j - t\beta_j \ge 0$. This is still a centered probability vector. Its cost is $C + t(\|Av_k\|^2 - C)$. Minimality forces $\|Av_k\|^2 \ge C$. Every original weight is positive and their costs average C, so every cost is exactly C. The second identity follows from $A = CS^{-1}$ and $\operatorname{tr} S^{-1} = \operatorname{tr} S/\det S$. □

Lemma 4. *A tight planar frame contains two indices with* $c^2 = (u_i^\top u_j/2)^2 \le 1/4$.

Proof. Write $u_i/\sqrt{2} = (\cos\theta_i, \sin\theta_i)$. Tightness implies that the unit-circle points $t_i = (\cos 2\theta_i, \sin 2\theta_i)$ have weighted mean zero. Reduce a convex representation of zero to at most three points. If two suffice, they are antipodal and the corresponding $c^2 = 0$. Otherwise their triangle contains zero in its interior; its three cyclic angular gaps are at most π and sum 2π. Some gap lies in $[2\pi/3, \pi]$, so that pair has $c^2 = (1 + \cos(2\theta_i - 2\theta_j))/2 \le 1/4$. A boundary triangle likewise has an antipodal pair. □

Proposition 5. *Every finite positive-energy residual frame satisfies* $\mathcal{T}_3 < 5E/6$.

Proof. Normalize $E = 1$ and suppose $\mathcal{T}_3 = 5/6$. Equation (7) forces $R = I_2/2$. In this case the dropped-correction anchor expression is

$$f(s) = s/2 + (1+s)/(4+2s), \qquad f(1) = 5/6, \quad f'(s) = 1/2 + 1/(2(s+2)^2) > 0.$$

Since $\mathcal{T}_3 \le f(\|w_i\|^2)$ for every i, all $\|w_i\|^2 \ge 1$. Their weighted mean is one and weights are positive, hence all equal one.

For every anchor, $H_i = I_2/2 + w_i w_i^\top/2$ has eigenvalues $1, 1/2$, so A_i is invertible and $\det H_i / \operatorname{tr} H_i = 1/3$. Equation (4) forces $\operatorname{tr}(A_i L_i A_i^\top) = 0$, hence $L_i = 0$ and $S_i = H_i$. The minimum squared norm of a two-point convex mean in this star must equal $1/3$: Lemma 2 bounds it above by $1/3$, while any smaller value would produce a certificate below $5/6$. Lemma 3 therefore forces, for every star neighbor,

$$\|H_i^{-1}\xi_{ij}\|^2 = \operatorname{tr}(H_i^{-1}) = 3. \tag{8}$$

Choose i, j from Lemma 4, and set $c = u_i^\top u_j/2$, $d = w_i^\top w_j$. This is an independent pair. In output coordinates with first axis w_i, the two components of $w_j - cw_i$ are $d - c$ and a number of squared magnitude $1 - d^2$. Also $b_{ij}^2 = 2(1-c^2)$, and H_i^{-1} has eigenvalues $1, 2$. Thus (8) is equivalent to

$$\frac{(d-c)^2 + 4(1-d^2)}{2(1-c^2)} = 3, \qquad 7c^2 - 2cd - 3d^2 = 2.$$

But exact square completion gives

$$7c^2 - 2cd - 3d^2 = \frac{22}{3}c^2 - 3(d + c/3)^2 \le \frac{11}{6} < 2,$$

a contradiction. This proves pointwise strictness. □

4 A uniform gap

Lemma 6. *For each positive-energy frame there is a residual frame on a subset of at most seven of its fixed atoms, with energy* $E' \le E$, *and with* $\mathcal{T}_3' \ge \mathcal{T}_3$. *If* $E' = 0$, *then* $\mathcal{T}_3 = 0$.

Proof. We recall Zhang's support reduction [3], Appendix F.1. Minimize $\sum_i q_i \|w_i\|^2$ over nonnegative weights on the fixed atoms satisfying the frame constraints. The symmetric input covariance gives three linear equations and the cross-covariance gives four. Normalization follows by taking the trace of the input equation, since $\|u_i\|^2 = 2$. Thus there are at most seven independent linear constraints. The feasible probability polytope is compact and nonempty. At a minimizing weight vector of least support, a dependence of its active constraint columns would permit both signs of small feasible perturbation. The energy slope must be zero by optimality. Continuing to the first zero weight would reduce support while preserving energy, a contradiction. The positive support consequently has at most seven atoms. Deleting atoms can only increase the selection minimum, giving $\mathcal{T}_3' \ge \mathcal{T}_3$. If $E' = 0$, all remaining w_i vanish; tightness still supplies an independent pair and its interpolator is zero. Then $\mathcal{T}_3' = 0$, so $\mathcal{T}_3 = 0$. □

Proof of the uniform frame assertion in Theorem 1. Let $\Gamma = \sup \mathcal{T}_3/E$ over all finite positive-energy frames. We know $\Gamma \le 5/6$. Suppose $\Gamma = 5/6$. Take a sequence of positive ratios tending to $5/6$, apply Lemma 6 (the reduced energy cannot be zero), and normalize each reduced energy to one. Its new ratio is at least the original ratio and at most $5/6$, so still tends to $5/6$. Write $\mathcal{T}_3^{(k)} \ge 5/6 - \epsilon_k$ with $\epsilon_k \downarrow 0$. There are at most seven atoms. Relabel and pass to a subsequence of constant support size, and diagonalize $R_k = \operatorname{diag}(r_k, 1 - r_k)$ by output rotations, which preserve all frame identities and costs.

By (7),

$$|\delta_k| = |2r_k - 1| \le \sqrt{60\epsilon_k/11} \longrightarrow 0.$$

For an arbitrary atom put $s_i^{(k)} = \|w_i^{(k)}\|^2$. The normalized anchor expression satisfies

$$g_{r_k}(a_i, b_i) \le f(s_i^{(k)}) + |\delta_k|/2.$$

This follows from $r_k(1-r_k) \le 1/4$ and $((1-r_k)a_i + r_k b_i)/2 \le (1+|\delta_k|)s_i^{(k)}/4$. As $\mathcal{T}_3^{(k)} \le g_{r_k}(a_i, b_i)$ for every atom and $f' \ge 1/2$,

$$s_i^{(k)} \ge 1 - \eta_k, \qquad \eta_k = 2\epsilon_k + |\delta_k| \longrightarrow 0.$$

Every summand below is nonnegative, and

$$\sum_i p_i^{(k)}(s_i^{(k)} - 1 + \eta_k) = \eta_k. \tag{9}$$

For every index set B, summing its nonnegative terms gives

$$\sum_{i\in B} p_i^{(k)} s_i^{(k)} \le (1-\eta_k)\sum_{i\in B} p_i^{(k)} + \eta_k \le \sum_{i\in B} p_i^{(k)} + \eta_k.$$

Thus atoms with vanishing probability cannot retain positive energy.

Pass to subsequences with $p_i^{(k)} \to p_i$ and $u_i^{(k)} \to u_i$; probabilities and input directions lie in compact sets. For $J = \{i : p_i > 0\}$, equation (9) forces $s_i^{(k)} \to 1$. These $w_i^{(k)}$ are bounded, so pass further to $w_i^{(k)} \to w_i$. For $i \notin J$, $p_i^{(k)} s_i^{(k)} \le p_i^{(k)} + \eta_k \to 0$. Its output covariance contribution tends to zero, and its cross-covariance has norm at most $\sqrt{2}\sqrt{p_i^{(k)}}\sqrt{p_i^{(k)} s_i^{(k)}} \to 0$. The surviving positive-weight atoms thus satisfy all frame identities, have unit output norm, total energy one, and covariance $R = I_2/2$.

By Proposition 5, this limiting frame has $\mathcal{T}_3^{(\infty)} < 5/6$. Take a witnessing convex combination of pair interpolators on at most three of its surviving indices. Every pair used has a nonzero limiting determinant; the same pairs are independent for all sufficiently large k, and their interpolators converge. Keeping its convex coefficients gives

$$\limsup_k \mathcal{T}_3^{(k)} \le \mathcal{T}_3^{(\infty)} < 5/6,$$

contradicting $\mathcal{T}_3^{(k)} \to 5/6$. Consequently $\Gamma < 5/6$. Set $\varepsilon_0 = 5/6 - \Gamma > 0$. This proves the uniform bound for positive energy. If $E = 0$, all w_i vanish and $\mathcal{T}_3 = 0$, so the same bound holds. No numerical lower bound on ε_0 is used. □

5 Return to regression

Let W° be the full-data minimum-norm minimizer and $\rho_i = W^\circ x_i - y_i$. The normal equations imply $\sum_i \rho_i x_i^\top = 0$, and for every B,

$$L_D(W^\circ + B) = L_D^* + N^{-1}\sum_i \|Bx_i\|^2.$$

Suppose $K = \sum_i x_i x_i^\top \succ 0$. Write $v_i = K^{-1/2}x_i$ and $\ell_i = \|v_i\|^2$. For $\ell_i > 0$, set

$$p_i = \ell_i/2, \qquad u_i = \sqrt{2/\ell_i}\, v_i, \qquad w_i = \sqrt{2/\ell_i}\, \rho_i.$$

The omitted indices have $x_i = 0$ and contribute only unavoidable loss. The remaining atoms satisfy (1), with $E = \sum_{\ell_i>0}\|\rho_i\|^2 \le NL_D^*$. For $\Theta = (W - W^\circ)K^{1/2}$, the excess loss is $\|\Theta\|_F^2/N$.

Given frame weights q, choose original weights proportional to q_i/ℓ_i. The selected objective, in Θ coordinates, is a positive multiple of $\sum_i q_i\|\Theta u_i + w_i\|^2$. If $U(q) = \sum_i q_i u_i u_i^\top \succ 0$, its unique minimizer is $\Theta = -G(q)$, where

$$G(q) = \left(\sum_i q_i w_i u_i^\top\right) U(q)^{-1} = \sum_{i<j:D_{ij}>0} \frac{q_i q_j D_{ij}}{\sum_{a<b} q_a q_b D_{ab}} G_{ij}, \qquad D_{ij} = \det(u_i, u_j)^2. \tag{10}$$

This follows by expanding the determinant and adjugate in dimension two: diagonal terms vanish and the i, j numerator term is $D_{ij}G_{ij}$.

The hulls defining $\mathcal{T}_3$ are closures of full-rank fits on the same support (cf. Zhang [3], Lemma E.1). For three pairwise independent directions and positive coefficients $\beta_{ij}, \beta_{ik}, \beta_{jk}$ of sum one, choose

$$q_i : q_j : q_k = D_{jk}/\beta_{jk} : D_{ik}/\beta_{ik} : D_{ij}/\beta_{ij}.$$

Substitution in (10) gives exactly the prescribed coefficients; boundary coefficients follow by limits. If two directions are parallel, the valid pairs share an anchor i. For a convex combination of its valid neighbors with coefficients α_j, take

$$q_i = 1 - t, \qquad q_j = t\frac{\alpha_j/D_{ij}}{\sum_k \alpha_k/D_{ik}}, \qquad 0 < t < 1.$$

Star-pair coefficients dominate the leaf pair as $t \downarrow 0$, so $G(q) \to \sum_j \alpha_j G_{ij}$. This also covers parallel leaves; endpoints use an independent pair. Every approximating fit is full rank. Consequently

$$\inf_{|\operatorname{supp} c| \le 3} L_D(\widehat{W}_c) \le L_D^* + \mathcal{T}_3/N \le (11/6 - \varepsilon_0)L_D^*.$$

No invariance of minimum-norm tie breaking under whitening is needed.

If the feature span has dimension one, write $x_i = a_i e$ and $W^\circ = \beta e^\top$. On nonzero a_i, the planar vectors ρ_i/a_i are centered with probabilities $a_i^2/\sum_j a_j^2$. Represent zero by at most three of them and choose original weights proportional to the representation coefficients divided by a_i^2. The selected normal equation recovers β; minimum norm sets the action on $e^\perp$ to zero. Rank zero is immediate. In the full-rank branch, $E = 0$ means that every nonzero-feature residual vanishes; a feature basis of size two then recovers W°. If $L_D^* = 0$, choose a basis of the feature span (at most two observations); its minimum-norm interpolator fits the full dataset. Thus all degenerate branches have ratio one, including the convention $0/0 = 1$. This proves Theorem 1. The lower bound $13/8$ is Zhang's Proposition 6.3 [3].

6 Formal verification and code

The companion Lean source is available at https://github.com/nZiben/weighted-selection-gap. It proves the scalar moment selection theorem, its anisotropy bound (7) in scalar form, optimality of the scalar relaxation, and the polynomial obstruction in Proposition 5. The selection theorem assumes only nonnegativity, normalization and the two scalar moments; it does not assume the desired selection conclusion.

This is a partial formalization. The planar lemma, equality rigidity, support reduction, compactness and the regression bridge are proved in this paper and are not formalized in Lean. In particular, the repository does not claim a Lean proof of Theorem 1. The source is checked using Lean 4.34.0 with a pinned Mathlib revision; reproduction commands and the compiler record are included. The audited declarations use only `propext`, `Classical.choice` and `Quot.sound`.

Use of AI

The proof was produced by GPT-6 Pro (GPT-6 Astra) in the ChatGPT user interface.

References

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